Novel NDVI-Based Vegetation Stress Analysis for Forest Fire Susceptibility Mapping in India

Mathematical Formulation · Physical Justification · Novel Contributions

Submitted to: Prof. Millie Pant

Department of Applied Mathematics & Scientific Computing

Indian Institute of Technology Roorkee

Data: MOD13A3 v061 | 1 km monthly NDVI | India | 2000–2025 (309 files)

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1 Background and Motivation

Gap in Biswas et al. (2025) Addressed by This Study

In **?**, NDVI is the **single most important variable** for Indian forest fire occurrence prediction: permutation importance = 22.3%, percentage contribution = 28.4% (highest among all 15 variables, Table 3 therein). Yet Biswas et al. applied NDVI as a single time-averaged composite map from MOD13C2 v006, with **no quality screening, no temporal decomposition, no anomaly extraction, and no pre-fire stress quantification.**

The present study extends **?** by applying five mathematically rigorous transformations to the 25-year monthly NDVI time series (MOD13A3 v061, 1 km, 309 monthly files, 2000–2025) to derive **seven novel NDVI-based features** (F1–F7) that are directly connected to the physics of fire ignition. Table 1 summarises all features and their novelty status.

Table 1: NDVI-derived feature set: this study vs **?**

ID	Feature	Core Equation	Physical Meaning	Novel?
F1	QA-filtered NDVI	Eq. (2)	Cloud-screened vegetation density	Partial
F2	Climatological mean $\bar{\mu}^{(m)}$	Eq. (14)	Seasonal baseline per month	No
F3	Monthly anomaly δ	Eq. (15)	Inter-annual vegetation stress	**Yes**
F4	STL trend $T(t)$	Eq. (4)	25-year vegetation change direction	**Yes**
F5	STL residual $R(t)$	Eq. (4)	Sudden stress: drought, fire-scar	**Yes**
F6	Mann–Kendall τ	Eq. (10)	Statistical significance of trend	**Yes**
F7	CVSI(k^*)	Eq. (16)	Cumulative pre-fire moisture deficit	**Yes**
Biswas et al. (2025): 1 feature				*This study: 7 features (5 novel)*

2 NDVI: Definition and Scale Correction

NDVI measures how green and healthy vegetation is. A healthy forest absorbs red light (for photosynthesis) and reflects near-infrared light. Dry, stressed, or burned vegetation does the opposite. High NDVI (> 0.6) = dense healthy forest. Low NDVI (< 0.2) = sparse, stressed, or burned vegetation.

Definition 2.1 (Normalized Difference Vegetation Index).

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}}} \tag{1}$$

where ρ_{NIR} and ρ_{Red} denote surface reflectance in the Near-Infrared and Red spectral *bands, respectively. The index ranges over* $[-0.2,\ 1.0]$.

MODIS MOD13A3 stores NDVI as signed 16-bit integers. The corrected floating-point NDVI for spatial pixel (i, j) at monthly time step t is:

Equation F1 — Scale Correction and Fill-Value Masking

$$\mathrm{NDVI}_{ij}^{(t)} = \begin{cases} \mathrm{raw}_{ij}^{(t)} \times 10^{-4} & \text{if } -2000 \le \mathrm{raw}_{ij}^{(t)} \le 10000 \\ \mathrm{NaN} & \text{otherwise} \end{cases} \tag{2}$$

Fill value: -3000 *(replace with NaN). Valid NDVI range after scaling:* $[-0.2,\ 1.0]$.

Quality Assurance (QA) Filtering

The `pixel_reliability` Science Data Set (SDS) from MOD13A3 provides a pixel-quality flag $q_{ij}^{(t)}$:

$$\mathrm{NDVI}_{ij}^{(t),\,\mathrm{QA}} = \begin{cases} \mathrm{NDVI}_{ij}^{(t)} & \text{if } q_{ij}^{(t)} \in \{0,\ 1\} \\ \mathrm{NaN} & \text{otherwise} \end{cases} \tag{3}$$

where $q = 0$ (Good data), $q = 1$ (Marginal data, usable), $q = 2$ (Snow/Ice — exclude), $q = 3$ (Cloudy — exclude), $q = -1$ (Fill/No-data). **?** applied **no** such quality screening.

3 Novel Contribution 1: STL Temporal Decomposition

Think of your 309 monthly NDVI images as a 25-year video of India's forests. Each pixel has a time series with 309 values. STL separates this into three stories: (1) Is the forest getting greener or browner over 25 years? [Trend] (2) What is the normal monsoon cycle? [Seasonal] (3) What unexpected events happened? [Residual — drought, fire-scars]

3.1 STL Decomposition Model

The STL (Seasonal and Trend decomposition using LOESS) algorithm (**?**) decomposes the NDVI time series at each pixel (i, j) additively:

Equation F4/F5 — STL Decomposition

$$\mathrm{NDVI}_{ij}^{(t)} = \underbrace{T_{ij}^{(t)}}_{\text{Trend}} + \underbrace{S_{ij}^{(t)}}_{\text{Seasonal}} + \underbrace{R_{ij}^{(t)}}_{\text{Residual}} \tag{4}$$

Applied independently to every spatial pixel. Period $p = 12$ months. Robust fitting: yes (downweights outliers). Time index $t = 1, 2, \dots, T$ where $T = 309$ monthly steps.

3.2 Component Definitions and Physical Interpretations

Trend Component $T_{ij}^{(t)}$ (Feature F4). The trend is estimated via a LOESS (Locally Estimated Scatterplot Smoothing) smoother. For a time point t_0, the LOESS estimate is obtained by solving the weighted local polynomial regression:

$$\hat{T}(t_0) = \arg\min_{\beta_0, \beta_1} \sum_{t=1}^{T} \mathcal{K}_h(t - t_0) \left[\mathrm{NDVI}_{ij}^{(t)} - \beta_0 - \beta_1(t - t_0)\right]^2 \tag{5}$$

where $\mathcal{K}_h$ is the tri-cube kernel weight function:

$$\mathcal{K}_h(u) = \left(1 - \left|\frac{u}{h}\right|^3\right)^3 \cdot \mathbf{1}[|u| < h] \tag{6}$$

and $h > 0$ is the bandwidth controlling smoothness.

Remark 3.1. *If $\partial T_{ij}/\partial t < 0$ over the study period, the pixel's vegetation is in long-term decline — indicating forest degradation, reduced canopy moisture, and elevated fire susceptibility.* **?** *cannot detect this from a single temporal average.*

Seasonal Component $S_{ij}^{(t)}$ (removed from analysis). The periodic annual monsoon cycle ($p = 12$ months) is isolated and *removed* to de-seasonalise the signal. This prevents normal dry-season dryness from being misclassified as genuine vegetation stress.

Residual Component $R_{ij}^{(t)}$ (Feature F5). After removing trend and seasonal components:

$$R_{ij}^{(t)} = \mathrm{NDVI}_{ij}^{(t)} - T_{ij}^{(t)} - S_{ij}^{(t)} \tag{7}$$

Large negative residuals ($R_{ij}^{(t)} \ll 0$) indicate sudden, unexpected vegetation drops — drought episodes, pest outbreaks, or post-fire regeneration patterns. These are the **pre-fire stress fingerprints** that a time-averaged NDVI completely conceals.

3.3 Mann–Kendall Trend Significance Test (Feature F6)

After computing the pixel-wise trend component $T_{ij}^{(t)}$, the non-parametric Mann–Kendall test (??) determines whether the observed trend is statistically significant or could arise by random chance.

3.3.1 Test Statistic

For a pixel time series $\{T^{(1)}, T^{(2)}, \ldots, T^{(n)}\}$ (where $n = T = 309$), define the Mann–Kendall statistic:

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^{n} \operatorname{sgn}\left(T^{(j)} - T^{(k)}\right) \tag{8}$$

where:

$$\operatorname{sgn}(x) = \begin{cases} +1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases} \tag{9}$$

3.3.2 Normalised Kendall's τ

Equation F6 — Mann–Kendall τ Statistic

$$\tau = \frac{P - Q}{\frac{1}{2}\, n\,(n-1)} \tag{10}$$

where P = number of concordant pairs (later value > earlier value), Q = number of discordant pairs (later value < earlier value), $\tau \in [-1, +1]$.

3.3.3 Variance and z-Score

For large n without ties:

$$\operatorname{Var}(S) = \frac{n(n-1)(2n+5)}{18} \tag{11}$$

The standardised test statistic is:

$$z = \begin{cases} \dfrac{S-1}{\sqrt{\operatorname{Var}(S)}} & \text{if } S > 0 \\ 0 & \text{if } S = 0 \\ \dfrac{S+1}{\sqrt{\operatorname{Var}(S)}} & \text{if } S < 0 \end{cases} \tag{12}$$

The two-tailed p-value is: $p = 2\,[1 - \Phi(|z|)]$ where Φ is the standard normal CDF.

3.3.4 Decision Rule and Spatial Output Maps

$$\text{Pixel classification} = \begin{cases} \textbf{Significant Browning} & \tau < 0 \text{ and } p < 0.05 \\ \textbf{Significant Greening} & \tau > 0 \text{ and } p < 0.05 \\ \textbf{No Reliable Trend} & p \geq 0.05 \end{cases} \tag{13}$$

The code produces two output maps:

- `browning_sig`: pixels with statistically significant vegetation decline ($\tau < 0$, $p < 0.05$) — these are the **highest fire-risk zones**.
- `greening_sig`: pixels with statistically significant vegetation recovery ($\tau > 0$, $p < 0.05$).

★ Novel Contribution: Novel Contribution 1 — STL Decomposition and Mann–Ke

- **?** used a single time-averaged NDVI value from MOD13C2 v006, with no temporal decomposition.
- This study applies pixel-wise STL to 309 monthly values (2000–2025), producing three physically distinct components: Trend T (Feature F4), Seasonal S (removed), Residual R (Feature F5).
- Mann–Kendall τ (Feature F6) confirms statistical significance of each browning/greening pixel.
- A pixel with $\tau = -0.31$, $p = 0.002$ in Uttarakhand (pauri Garhwal, 30.2°N, 78.8°E) represents a net NDVI loss of 0.075 over 25 years — invisible to a single-value approach.
- **References: ?**; **?**; **?**.

4 Novel Contribution 2: Monthly NDVI Anomaly

Indian forests are always greener in the monsoon (July–September) and browner in pre-monsoon (March–May). If you look at raw NDVI in May, every forest looks stressed — but that is *normal*. The anomaly δ strips this away and asks: "Is this May more stressed than the average May over 20 years?" A negative δ means genuine abnormal stress — the real fire-relevant signal.

4.1 Climatological Monthly Mean (Feature F2)

Using the 2001–2020 baseline (consistent with **?**):

Equation F2 — Climatological Mean

$$\bar{\mu}_{ij}^{(m)} = \frac{1}{|\mathcal{Y}_m|} \sum_{y \in \mathcal{Y}_m} \mathrm{NDVI}_{ij}^{(y,m),\,\mathrm{QA}} \tag{14}$$

where $m \in \{1, 2, \ldots, 12\}$ is the calendar month, $\mathcal{Y}_m = \{2001, 2002, \ldots, 2020\}$, and only non-`NaN` values contribute. This produces **12 mean maps** — one per calendar month.

4.2 Monthly NDVI Anomaly (Feature F3)

Equation F3 — NDVI Anomaly δ

$$\boxed{\delta_{ij}^{(y,m)} = \mathrm{NDVI}_{ij}^{(y,m),\,\mathrm{QA}} - \bar{\mu}_{ij}^{(m)}} \tag{15}$$

- $\delta_{ij}^{(y,m)} < 0$: below-average greenness (**vegetation stress**) — fire risk **increases**.
- $\delta_{ij}^{(y,m)} > 0$: above-average greenness — fire risk decreases.
- By construction: $\sum_{y \in \mathcal{Y}_m} \delta_{ij}^{(y,m)} \approx 0$ (zero mean over baseline).

Proposition 4.1 (Anomaly as De-seasonalised Signal)**.** *The anomaly $\delta_{ij}^{(y,m)}$ removes the confounding seasonal cycle and exposes the inter-annual vegetation stress signal that is causally connected to fire ignition, independent of the time of year.*

★ Novel Contribution: Novel Contribution 2 — Monthly NDVI Anomaly

- **?** cannot distinguish between "normal dry season" and "abnormally stressed vegetation."
- The anomaly $\delta_{ij}^{(y,m)}$ isolates genuine inter-annual stress signals from the expected monsoon cycle.
- **Concrete example — Uttarakhand, March 2016 (El Niño year):** Raw NDVI $= 0.38$ (looks the same as any dry March). Anomaly $\delta = 0.38 - 0.44 = -0.06$ — the forest is 6 NDVI-units below its 20-year March average. This is the genuine drought stress signal. **?** cannot see this.
- This anomaly becomes the direct input to the CVSI (Section 5) and Feature F3 in the ML pipeline.

5 Novel Contribution 3: Cumulative Vegetation Stress Index (CVSI)

Consider a forest in Odisha. In January: NDVI slightly below normal ($\delta = -0.03$). In February: still below normal ($\delta = -0.05$). In March: again below ($\delta = -0.04$). In April: **fire breaks out.**

Biswas et al. see only: April NDVI = 0.35. They cannot see that three months of accumulated stress preceded the fire.
The CVSI adds up all negative anomalies in the k months *before* each fire event — quantifying the multi-month moisture deficit that enabled ignition.

5.1 Mathematical Definition

Let τ_f denote the calendar month of fire occurrence at pixel (i, j). The k-month Cumulative Vegetation Stress Index is:

Equation F7 — Cumulative Vegetation Stress Index (CVSI)

$$\boxed{\mathrm{CVSI}_{ij}(\tau_f,\, k) = \sum_{t=\tau_f-k}^{\tau_f-1} \max\left(-\delta_{ij}^{(t)},\, 0\right)} \tag{16}$$

The operator $\max(-\delta, 0)$ retains **only** stress episodes (negative anomaly months) and discards above-average greenness months. $k \in \{1, 2, 3, 4, 5, 6\}$ is the accumulation window (months).

5.2 Expanded Form for $k = 3$

$$\mathrm{CVSI}_{ij}(\tau_f,\, 3) = \underbrace{\max\left(-\delta_{ij}^{(\tau_f-3)},\, 0\right)}_{\text{3 months before}} + \underbrace{\max\left(-\delta_{ij}^{(\tau_f-2)},\, 0\right)}_{\text{2 months before}} + \underbrace{\max\left(-\delta_{ij}^{(\tau_f-1)},\, 0\right)}_{\text{1 month before}} \tag{17}$$

Each term is zero if that month had above-average greenness, and positive if there was stress. The sum accumulates the total moisture deficit in the pre-fire window.

5.3 Optimal Lag Selection via Mutual Information

The optimal accumulation window k^* is selected by maximising the Shannon mutual information between the CVSI and binary fire occurrence labels $F_{ij} \in \{0, 1\}$:

$$k^* = \underset{k \in \{1,2,3,4,5,6\}}{\arg\max} \; I(\mathrm{CVSI}(\tau_f,\, k)\; ;\; F_{ij}) \tag{18}$$

where the mutual information is:

$$I(X;\, Y) = \sum_{x,\, y} p(x, y)\, \log_2 \frac{p(x, y)}{p(x)\, p(y)} \tag{19}$$

This information-theoretic criterion identifies which lag window most reduces uncertainty about fire occurrence — a rigorous alternative to the fixed-window approaches used in the literature.

5.4 Physical Justification

Proposition 5.1 (CVSI as Fuel Moisture Deficit Proxy)**.** *Prolonged below-average NDVI ($\delta < 0$) signals:*

1. *Reduced canopy moisture content (leaf water potential decreasing),*
2. *Accelerated fine fuel drying (dead leaf litter moisture $\downarrow$),*
3. *Increased bulk density of surface fuels susceptible to ignition.*

The CVSI therefore serves as a proxy for the ***pre-fire fuel moisture deficit*** *— a physically established precursor to wildfire ignition (??).*

★ Novel Contribution: Novel Contribution 3 — Cumulative Vegetation Stress Inde

- No prior Indian national-scale forest fire study has introduced a temporally accumulating pre-fire NDVI stress metric.
- **?** use NDVI at the analysis date only — they have no memory of preceding months.
- CVSI quantifies multi-month fuel moisture deficit before ignition: a physically grounded causal feature.
- Optimal lag k^* is selected via mutual information (Eq. 18) — a rigorous information-theoretic criterion absent from all prior Indian fire studies.
- **References:** ?; ?.

6 Feature Engineering Summary and Integration

Table 2 presents all seven NDVI-derived features with their complete mathematical definitions.

Table 2: Complete mathematical summary of NDVI-derived features

ID	Feature	Defining Equation	Novel?
F1	QA NDVI	$\mathrm{NDVI}_{ij}^{(t)} = \mathrm{raw}_{ij}^{(t)} \times 10^{-4}$ if $q_{ij}^{(t)} \in \{0,1\}$; else NaN	Partial
F2	Climatological mean	$\bar{\mu}_{ij}^{(m)} = \frac{1}{\lvert\mathcal{Y}_m\rvert} \sum_{y\in\mathcal{Y}_m} \mathrm{NDVI}_{ij}^{(y,m)}$	No
F3	Anomaly δ	$\delta_{ij}^{(y,m)} = \mathrm{NDVI}_{ij}^{(y,m)} - \bar{\mu}_{ij}^{(m)}$	**Yes**
F4	STL Trend $T(t)$	$\mathrm{NDVI}_{ij}^{(t)} = T_{ij}^{(t)} + S_{ij}^{(t)} + R_{ij}^{(t)}$; LOESS smoother with period $p = 12$	**Yes**
F5	STL Residual $R(t)$	$R_{ij}^{(t)} = \mathrm{NDVI}_{ij}^{(t)} - T_{ij}^{(t)} - S_{ij}^{(t)}$	**Yes**
F6	M-K τ	$\tau = (P - Q) \,/\, \frac{1}{2}n(n-1)$; sig. if $p < 0.05$	**Yes**
F7	CVSI(k^*)	$\mathrm{CVSI}_{ij}(\tau_f, k^*) = \sum_{t=\tau_f-k^*}^{\tau_f-1} \max(-\delta_{ij}^{(t)}, 0)$	**Yes**

These seven features replace the single time-averaged NDVI of **?** and expand the predictor space from 15 to 21 dimensions in the ML pipeline (RF, XGBoost, SVM + MaxEnt). SHAP (SHapley Additive exPlanations) analysis (**?**) is applied post-training to decompose each feature's marginal contribution.

6.1 Concrete Validation: Uttarakhand Fire Pixel

Table 3 compares what **?** see versus what this study extracts for a single fire pixel in Pauri Garhwal district, Uttarakhand (30.2°N, 78.8°E) — a confirmed high-fire zone from MODIS FIRMS data (April 2016).

Table 3: Information extracted per methodology — Uttarakhand fire pixel (April 2016)

Information	Biswas et al. (2025)	This Study
NDVI input	0.45 (single 20-year average)	309 monthly values decomposed
Long-term trend	Not detectable	T: -0.003 NDVI/year; net loss $= 0.075$
Trend significance	Cannot test	$\tau = -0.31$, $p = 0.002$ — significant browning
Pre-fire anomalies	Not detectable	$\delta_{\text{Jan}} = -0.03$, $\delta_{\text{Feb}} = -0.05$, $\delta_{\text{Mar}} = -0.04$
CVSI ($k = 3$)	Not available	$0.03 + 0.05 + 0.04 = 0.12$ (high stress)
Model verdict	*Moderate risk*	**HIGH FIRE RISK**

7 Conclusion

This technical note has presented five novel extensions to the NDVI preprocessing methodology of **?**, summarised as follows:

NC1. STL Decomposition (Eqs. 4–7): pixel-wise separation of NDVI into trend T, seasonal S, and residual R components over 309 monthly time steps.

NC2. Mann–Kendall Test (Eqs. 10–13): non-parametric statistical validation of browning/greening trends at the $\alpha = 0.05$ level.

NC3. Monthly Anomaly (Eq. 15): removal of the seasonal cycle to expose inter-annual vegetation stress independent of time of year.

NC4. CVSI (Eq. 16): a novel multi-month pre-fire fuel moisture deficit index with information-theoretically optimal lag selection.

NC5. QA Filtering (Eq. 3): pixel-level cloud and snow screening absent in **?**.

Together these transform 1 raw NDVI feature into 7 physically interpretable, temporally dynamic predictors (F1–F7), expanding the ML feature space from 15 to 21 dimensions relative to the baseline study. The methodology is applicable at the national scale for India's full claimed territory using 309 monthly MOD13A3 v061 files (March 2000 – December 2025).